controls all its own identities, participates in a federated ecosystem, or allows users to manage their own decentralized identifiers, each approach shapes how identities are created, authenticated, discovered, and trusted.

Identity system models—centralized, federated, and decentralized—each balance control, interoperability, and security differently, impacting how identity is managed and used for authorization. The model choice also impacts the quality and timeliness of identity attributes, which in turn directly affects authorization decisions. In the sections that follow, we'll examine the strengths, limitations, and implications of each model, focusing on how they influence dynamic authorization.

2.5.1 Centralized identity: single authority, high control

The centralized identity model is the one most of us think of when we think of an identity system. The identity system your employer uses to manage workforce identity is centralized, and so is your state-issued driver's license or ID. Even the loyalty program at the local supermarket is a centralized identity system. These systems are controlled by a single entity that provides them and determines how they are used.

Most organizations no longer build their own identity systems, opting to purchase one instead. These are often cloud services, operated on behalf of the organization. They are usually architected for a specific purpose, like managing employees.

One advantage of the centralized identity model is strong control and enforcement. Because the organization controls the system, they get to make the rules. This allows the organization to tailor the identity system to their specific needs. The centralized model also simplifies identity management since governance is kept within the organization. From the standpoint of authorization, the control that centralized models provide means that they can usually be more easily configured to provide the attributes needed for authorization decisions.

The biggest risk in the centralized model is that it represents a single point of failure—carefully choosing the system provider is critical to avoid failures that would cripple the business.